

Nicholas Campbell

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Education

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|-------------------------|--|-------------|
| MEng (Cum Laude) | Stellenbosch University , Mechatronic Engineering | 2024 – 2025 |
| | <ul style="list-style-type: none">• Thesis: An Architecture for a Federated Digital Twin System for an Automotive Enterprise (80%)• Conducted in close collaboration with BMW IT Hub South Africa• Presented conference paper at SOHOMA 2025, Aix-en-Provence, France (Oct 2025) | |
| BEng | Stellenbosch University , Mechatronic Engineering | 2020 – 2023 |
| | <ul style="list-style-type: none">• Best Final Year Biomedical Project — Institute of Biomedical Engineering (Wearable Cardiopulmonary Exercise System, 90%), Dec 2023 | |

Experience

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|---|--------------------------|
| SpatialEdge , Data Engineer | 2026 – present
1 year |
| <ul style="list-style-type: none">• Currently migrating on-premises data infrastructure to a cloud-native architecture on GCP, creating a centralised lake-house for data analytics and ML use cases.• Migrated legacy data infrastructure to a cloud-native architecture on GCP, ingesting on-premises relational databases, SaaS platforms (Salesforce, Jira), ERP systems, and internal case management systems into BigQuery• Architected and deployed an automated Salesforce → BigQuery ingestion pipeline using Python and BigQuery Data Transfer Service; built a CLI utility that auto-generates YAML configs from the Salesforce API, reducing new-asset onboarding from hours to minutes• Engineered BigQuery optimisations — transforming standard views into sharded tables and using table cloning to provide high-performance writable environments without additional storage costs• Automated GitLab CI/CD pipelines to streamline deployment cycles and synchronise configurations with cloud infrastructure, incorporating dual-stage validation and POPIA compliance tagging• Led production deployments and participated fully in Agile sprint ceremonies | |
| BMW IT Hub South Africa , Graduate Researcher | 2024 – 2025
1 year |
| <ul style="list-style-type: none">• Collaborated with multiple stakeholders to derive business requirements, translating them into technical requirements for a system.• Designed software architectures integrating heterogeneous digital twin platforms across the vehicle lifecycle, applying cloud services and knowledge graphs for DevOps integration and lifecycle analytics.• Participated in the BMW Automotive IT Colloquium (Jun 2025) on AI, Digital Twins, and Automation; contributed to a hackathon building a VS Code extension that integrates an LLM with the GitHub API to enhance developer efficiency across the DevOps lifecycle | |
| Stellenbosch University , Head Teaching Assistant — Mechatronics 424 | 2025 – 2025
1 year |
| <ul style="list-style-type: none">• Oversaw execution of Mechatronics 424 group projects; coordinated demonstrators, managing tasks, timelines, and team communication | |

Zutari, Engineering Intern

2022 – 2023

- Rotated through Industrial Automation and Water teams
- Developed a GeoFencing application in Ignition Perspective using phone-based location data to demonstrate safety and control strategies

1 year

Projects

Personal Finance Tracker

2026

- Full-stack personal finance dashboard built with Next.js (TypeScript), Supabase (PostgreSQL), and deployed on Vercel; ingests bank transactions nightly via a Gmail Apps Script integration
- Engineered a nightly ETL pipeline (raw → prepared) with multi-currency FX conversion, a three-pass categorisation engine (rules → LLM → manual overrides), and Vercel Cron Jobs for scheduling
- Features net worth tracking, budget management, recurring transaction detection, and a GitHub-style spend heatmap across 8 dashboard pages

Wearable Cardiopulmonary Exercise System

2023

- Designed, built, and tested a low-cost mobile CPET system with wireless real-time data streaming up to 90 m; strong correlation with reference system
- Awarded Best Final Year Biomedical Project by the Institute of Biomedical Engineering (90%)

GeoFencing Application

2023

- Built a concept demonstrator in Ignition Perspective during a vacation-work period at Zutari, using phone-based GPS data to implement safety and control strategies for the Industrial Automation team

Analog Audio Amplifier

2023

- Designed, built, and tested a class AB 2-channel audio amplifier outputting 8W as part of Electronics 315 (undergraduate mechatronics)

Publications

An Architecture for a Federated Digital Twin System for an Automotive Enterprise

May 2026

Nicholas Campbell, Prof. Anton H. Basson, Prof. Karel K. Kruger

[10.1007/978-3-032-18786-4_15](#) (Proceedings of SOHOMA (Service Oriented, Holonic and Multi-agent Manufacturing Systems for Industry of the Future) 2025)

Skills

Languages: Python, SQL, JavaScript, C

Databases: BigQuery, PostgreSQL, Oracle, MySQL

Cloud & Infrastructure: Google Cloud Platform (BQ, GCS, Dataproc), Apache Airflow, GitLab CI/CD, GitHub Actions, Docker; some experience with AWS & Azure

Data Engineering: ETL pipeline development, data warehouse optimisation, CI/CD automation, Spark

Mechatronics: Embedded systems, CAD, control systems, signal processing

Domain Knowledge: Systems integration, software architectures for data aggregation, digital twins, knowledge graphs

Professional: Agile methodology, sprint planning, cross-functional collaboration, technical documentation, production deployment management

Awards

Best Final Year Biomedical Project: Institute of Biomedical Engineering, Dec 2023 — Wearable Cardiopulmonary Exercise System (90%)

Data School Hackathon — 2nd Place: Standard Bank & Mobalyz, Aug 2023 — real-time driver risk-scoring system

Golden Key Society: Invited 2021, 2022, 2023